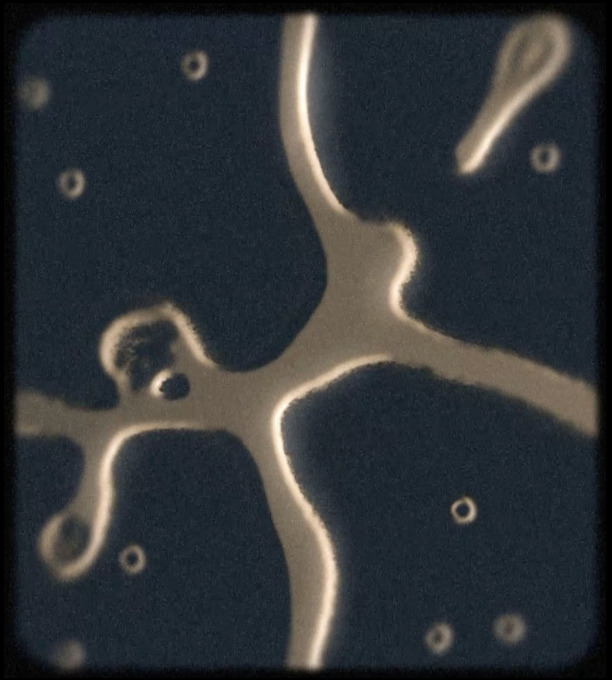
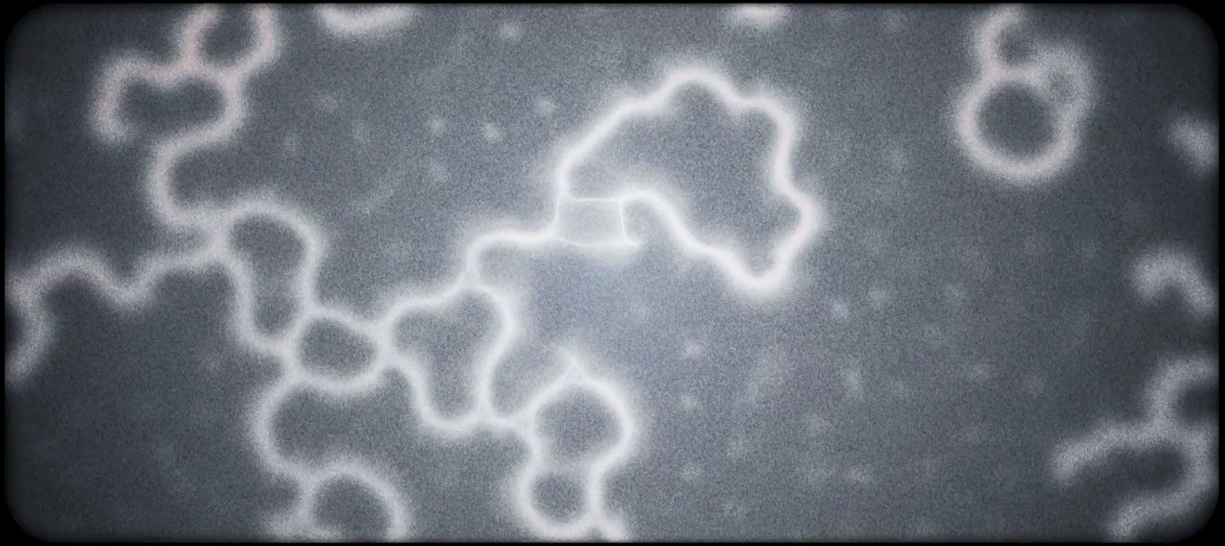
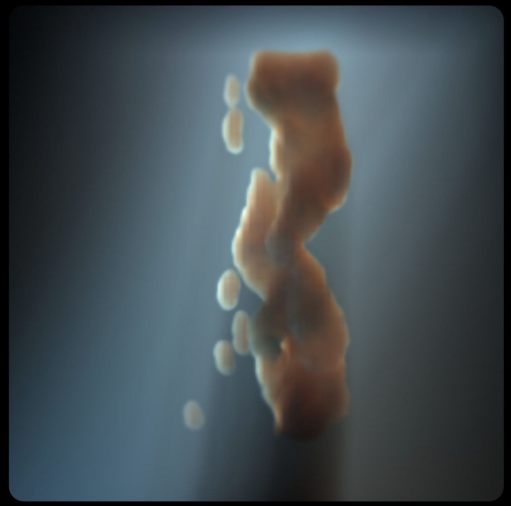
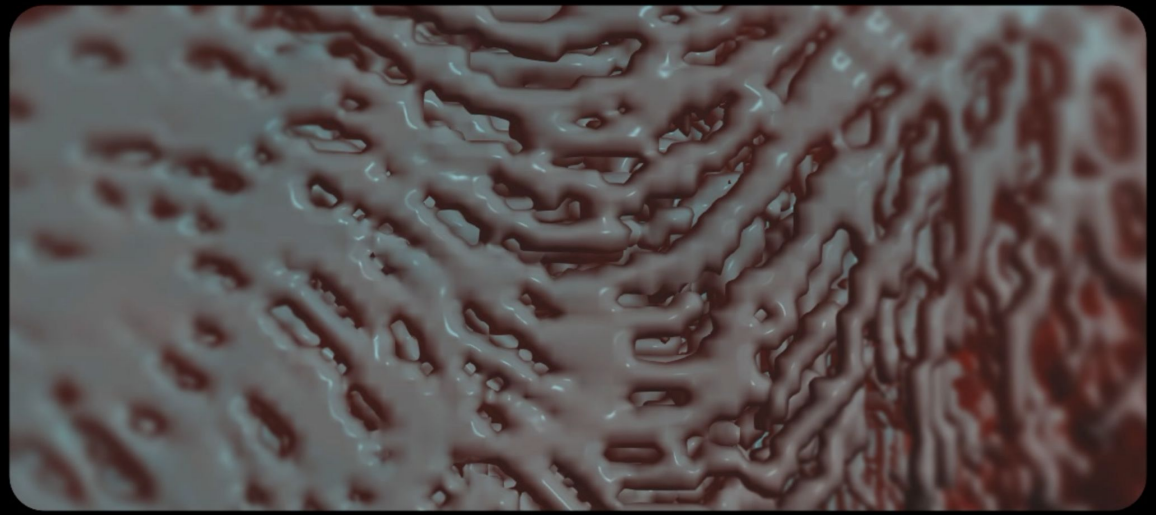
MICROCOSM

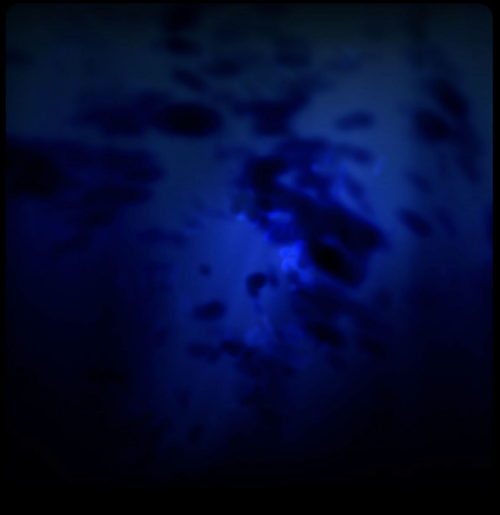
Xavier Cox

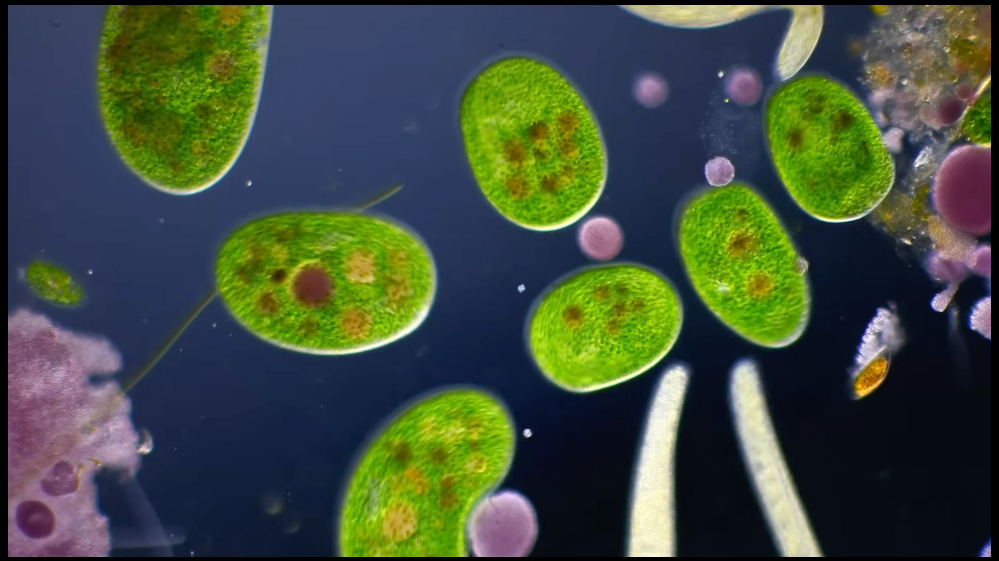










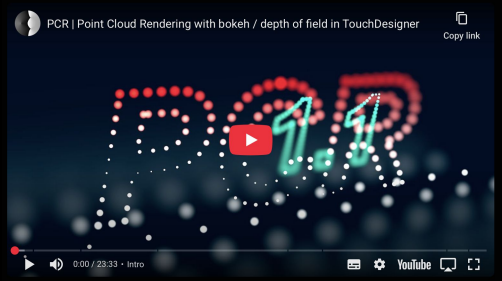


Josef Pelz

7.3k members · 53 posts

Hi! I'm Josef Pelz, a digital artist based in Berlin. In my work I explore the emergent behavior of generative systems. My practice is strongly influenced by my studies of mathematics and computer science. TouchDesigner is my tool of choice, where I...

Gift membership

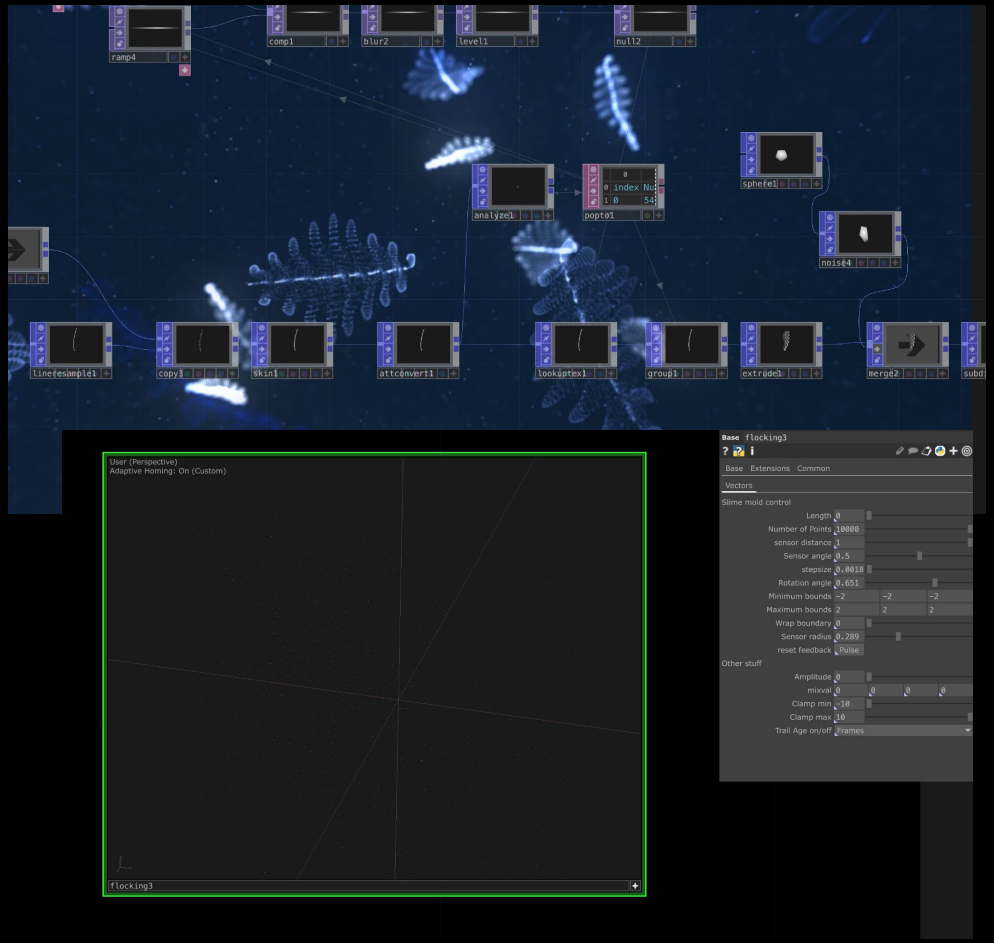


PCR | Tutorial (Version 1.1)

Josef Pelz
May 13

8 3

In this tutorial I explain how to use the Point Cloud Rendering (PCR) tool in TouchDesigner. PCR allows you to render point clouds very efficiently (when using the pixel mode on Windows) and with depth of field.


ramp4
comp1
blur2
level1
null2
analyze1
popto1
sphere1
noise4
linearesample1
copy3
skin1
attconvert1
lookuptex1
group1
extrude1
merge2
User (Perspective)
Adaptive Homing: On (Custom)
flocking3
Base flocking3
Base Extensions Common
Vectors
Slime mold control
Length 0
Number of Points 10000
sensor distance 1
Sensor angle 0.5
stepsize 0.0018
Rotation angle 0.651
Minimum bounds -2 -2 -2
Maximum bounds 2 2 2
Wrap boundary 0
Sensor radius 0.289
reset feedback Pulse
Other stuff
Amplitude 0
mixval 0 0 0 0
Clamp min -10
Clamp max 10
Trail Age on/off Frames

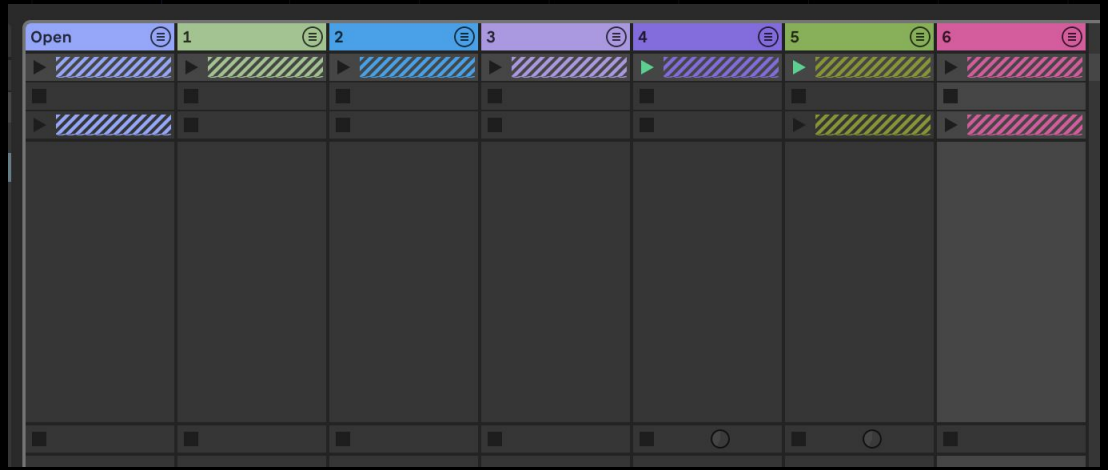
transform6
PCR_POP
transform7
PCR_POP2
transform11
PCR_POP5

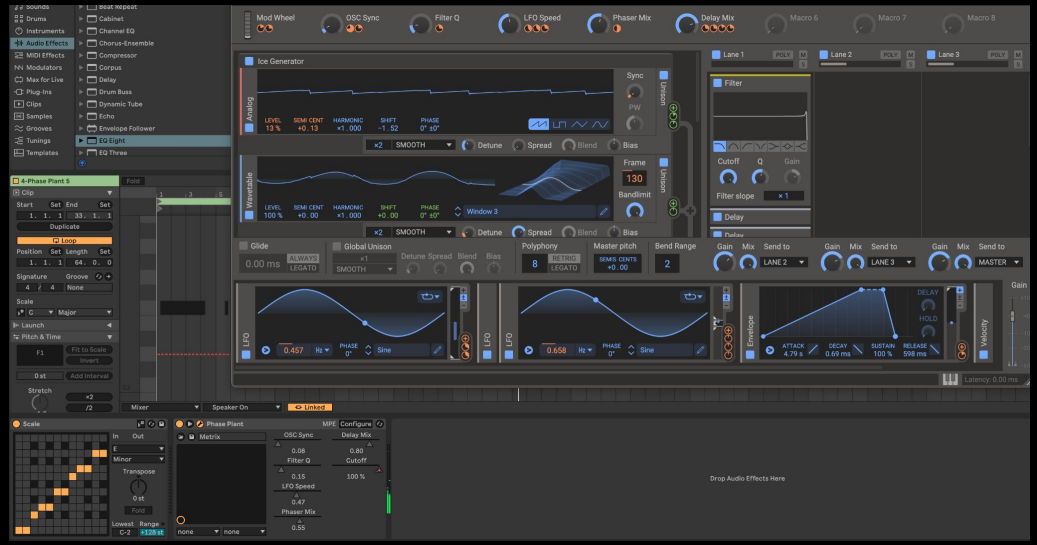
```python
# me  - this DAT
# frame - the current frame
# state - True if the timeline is paused
#
# Assumptions:
# - Containers exist in the same network as this DAT and are named: one, two, three, four
# - A Switch TOP named switch1 selects inputs 0-3 for those containers

CONTAINER_NAMES = ['open', 'one', 'two', 'three', 'four', 'five', 'six']
SWITCH_PATH = 'switch1'  # use full path if not a sibling, e.g. '/project1/switch1'

def _active_idx():
    s = op(SWITCH_PATH)
    if not s:
        debug(f'Execute DAT: switch not found: {SWITCH_PATH}')
        return 0
    try:
        idx = int(round(float(s.par.index.eval())))
    except Exception as e:
        debug(f'Execute DAT: could not read switch index: {e}')
```

execute1

```python
# Execute DAT: Fire/Stop Ableton clip slots based on switch1 index (0..3)

# -------- Configuration --------
DEBUG = False

ROOT = me.parent()          # Where switch1 + abletonClipSlots live
SWITCH_OP = 'switch1'       # Switch TOP name
# Index -> abletonClipSlot COMP name mapping
INDEX_TO_SLOT = {
    0: 'Opener',
    1: 'First',
    2: 'Second',
    3: 'Third',
    4: 'Fourth',
    5: 'Fifth',
    6: 'Sixth',
}

# Common internal names for the two pulse parameters (label is "Fire Clip"/"Stop All Cli
FIRE_PARAM_CANDIDATES     = ['FireClip', 'Fireclip', 'Fire', 'TriggerClip', 'Trigger']
STOP_ALL_PARAM_CANDIDATES = ['StopAllClips', 'Stopallclips', 'StopAll', 'Stop']
```

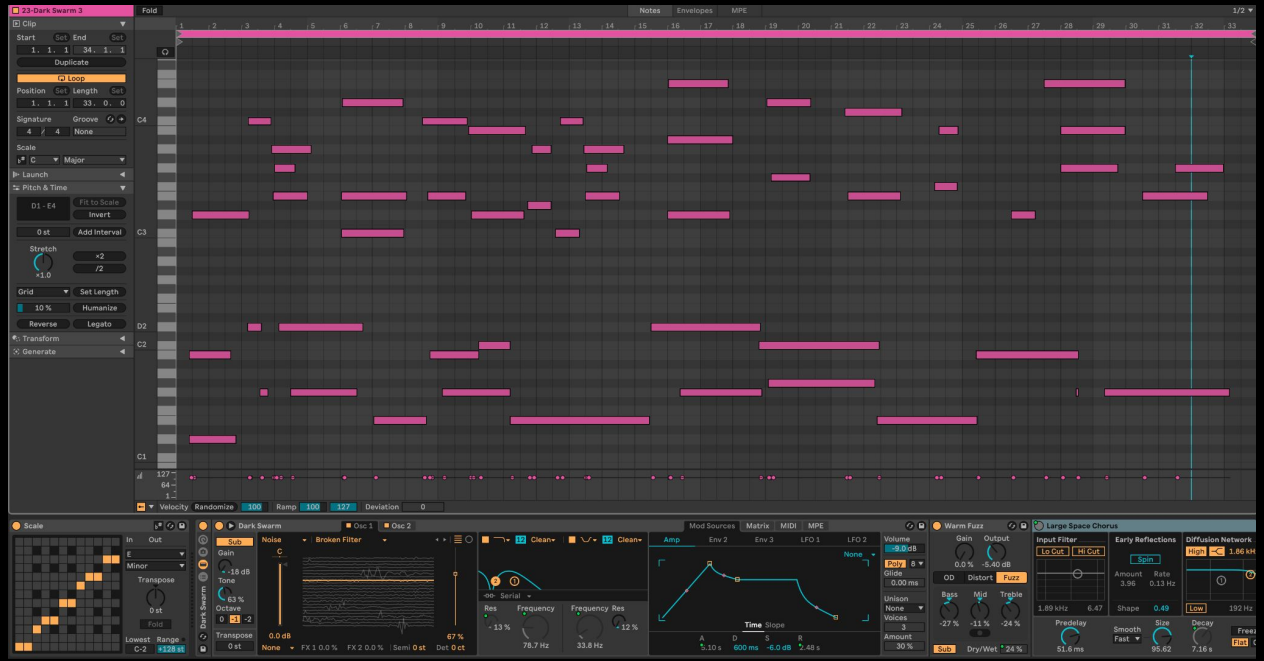
execute2




Mod Wheel
OSC Sync
Filter Q
LFO Speed
Phaser Mix
Delay Mix
Macro 6
Macro 7
Macro 8
Ice Generator
Wavetable
Filter
Delay
Drop Audio Effects Here

4-Audio 2
4-Audio 0002 [2025-10-16 131734].aif 44.1 kHz - 24-Bit - 2 Ch
Overdrive
Auto Filter
Longest Ping Pong
Drop Audio Effects Here

Osc Mix
Filter
Envelopes
LFO
Mod
Osc 1
-14 dB
Osc 2
0.0 dB
Noise
-10 dB
Freq
20.0 kHz
Type
II
Key
1.00
Res
0.47
HP
10.0 Hz
Freq Mod
LFO
Mod
100 %
100 %
1 Attack
0.00 ms
Decay
600 ms
Sustain
70 %
Release
12.7 ms
2 Attack
0.00 ms
Decay
32.1 ms
Sustain
100 %
Release
600 ms
Sync
1/12
Hz
1:1
ms
S&H
Amount
27 %
Mod
Mod
-16 %
Mode
Stereo
31 %
Voices
4
Drift
31 %
Legato
Glide
0.00 ms
Volume
-6.0 dB
Vel > Vol
32 %
Transpose
0 st
Note PB
PB Range
12 st
Channel EQ
Drum Buss
Bass Mono
Hybrid Reverb
Hybrid Reverb-2
Redux
Compressor
Multiband Dynamics
Limiter
TDA_Mapper
TDA Mapper
OSC Name
TDA_Mapper
Out Address
127.0.0.1
Out Port
58811
/map/t1
/TDA_Mapper
/map#
Hi Pass Filter
Map
TD
Unmap
Min
Max


23-Dark Swarm 3
Fold
Notes
Envelopes
MPE
1/2
Clip
Start
End
Duplicate
Loop
Position
Length
Signature
Groove
Scale
Major
Launch
Pitch & Time
Fit to Scale
Invert
Add Interval
Stretch
Grid
Set Length
Humanize
Reverse
Legato
Transform
Generate
C4
C3
D2
C2
C1
Velocity
Randomize
Ramp
Deviation
Scale
Transpose
Lowest
Range
Dark Swarm
Osc 1
Osc 2
Noise
Broken Filter
Mod Sources
Matrix
MIDI
MPE
Warm Fuzz
Large Space Chorus
Input Filter
Early Reflections
Diffusion Network
Predelay
Size
Decay

EQ Eight
Freq
8.00 kHz
Gain
1.00 dB
Q
0.33
Mode
Stereo
Edit
Adapt. Q
Off
Scale
100 %
Gain
0.00 dB
Glue Compressor
Attack
Release
Ratio
Threshold
Makeup
-23.6 dB
7.87 dB
Clip
Soft
Range
70.0 dB
Dry/Wet
50 %
Overdrive
557 Hz
7.94
Drive
Tone
0 %
60 %
Dynamics
20 %
Dry/Wet
20 %
Utility
Input
Stereo
Mid/Side
40S
Mono
Bass Mono
120 Hz
Output
Gain
4.00 dB
Balance
Mute
DC
Limiter
Gain
0.00 dB
Lookahead
6 ms
Release
300 ms
Auto
Ceiling
-0.30 dB
Stereo
Spectrum
Block
4096
Channel
Refresh
40.0 ms
Avg
Graph
Line
Max
Scale X
Lin
Log
ST
Range

MICROCOSM - Specimen #953